National Taiwan University **Advanced Structural Theory**
Department of Civil Engineering Instructor: C.-Y. Ke
Fall 2026

Programming Assignment 2
Issued: October 15, 2026 Due: November 5, 2026

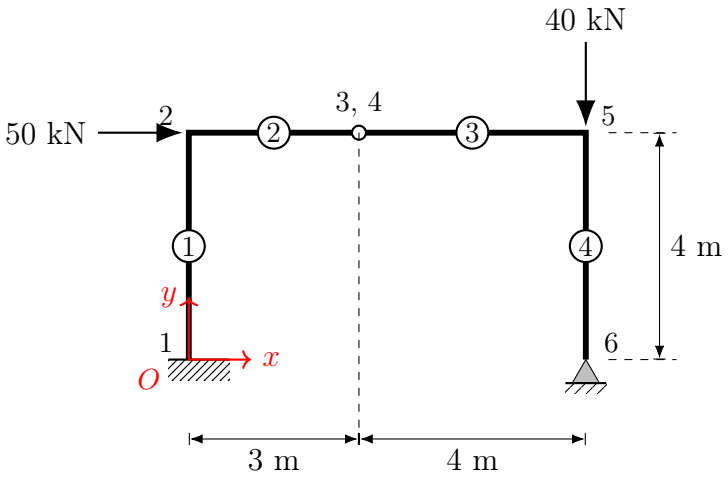
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1. Follow the instructions in `prog2.m` and extend your `FRAME26` from Programming Assignment 1: complete the functions `IDMAT`, `MEMDOF` and `SEMIBAND` (the function `LOAD` is given), and complete the main program up to the mark

   ```
   % ^^ UP TO HERE --- PROG 2 ^^
   ```

2. Run your program with the three input files of Programming Assignment 1 (the beam, the truss and the frame). For each file, print `IDND`, `NEQ`, `LM`, `NSBAND` and `GLOAD` in the MATLAB command window.

3. Prepare an input file for the 2D frame shown below using units of kN and m, and save it as `prog-2-2D_Frame.ipt`. The beam has an internal hinge at 3 m from the left column. Model the hinge with the double-node technique discussed in class: nodes 3 and 4 have the same coordinates, and in `NFIX` the translations of node 4 refer to node 3 (enter the node number 3) while its rotation is free. Node 1 is fixed and node 6 is a pin. Run your program with this file and print `IDND`, `NEQ`, `LM`, `NSBAND` and `GLOAD`.



Members	E (GPa)	ν	A (mm^2)	I (mm^4)
1, 4 (columns)	200	0.30	2×10^4	5×10^8
2, 3 (beam)	200	0.30	1.5×10^4	3×10^8

4. The semi-bandwidth `NSBAND` depends on how the nodes are numbered. Take the truss of Programming Assignment 1 (file `prog-1-2D_Truss.ipt`) and renumber its nodes so that the bottom-chord nodes are numbered 1 to 7 from left to right and the top-chord nodes 8 to 12 from left to right; keep the element numbers, the connectivity, the supports, the loads and the properties unchanged. Save the file as `prog-2-2D_Truss_renumbered.ipt` and run your program with it. In a text file `README.txt`, report `NSBAND` for the two numberings, explain in a few sentences why they differ, and state how much storage the banded global stiffness matrix `GLK(NEQ,NSBAND)` needs in each case.